Dalton Transactions

PAPER

Cite this: *Dalton Trans.*, 2025, **54**, 15333

A Ce-modified HY zeolite promotes polyethylene terephthalate hydrolysis and hydrogenation

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Received 9th May 2025,
Accepted 12th September 2025

DOI: 10.1039/d5dt01089f

rsc.li/dalton

The environmental problems arising from excessive polyethylene terephthalate (PET) production can be tackled through advanced recycling processes. Nevertheless, the stringent reaction conditions of high pressure and temperature pose obstacles to the advancement of these processes. Herein, we utilized an ion-exchange method to introduce Ce species into a HY zeolite and hydrolyzed PET under neutral conditions at 190 °C for 3 h, achieving a 93% TPA yield, which is 2.3 times higher than that of HY. Mechanism studies show that the incorporation of Ce species plays a crucial role in accelerating the generation of acid sites, which is essential for PET degradation. Additionally, within the composite system of Ce-HY and Pd, the Ce species play a crucial role in substantially boosting the H-spillover effect, which in turn promotes the hydrogenation of decomposition products. This research offers a straightforward and efficient approach for upgrading and recycling PET.

Introduction

Polyethylene terephthalate (PET) holds a prominent position among the most commonly used petroleum-derived plastics, being widely employed in food packaging, engineering plastics, and synthetic fibers.[1–3] Nevertheless, its resistance to natural degradation leads to the accumulation of substantial solid waste, posing a risk to the ecological environment.[4] Consequently, this ubiquitous problem highlights the importance of advanced recycling strategies. In comparison with mechanical recovery, chemical depolymerisation allows for better structural retention for upgrading to more high-value chemicals. Most chemical depolymerization methods (ammonolysis, alcoholysis, hydrolysis, *etc.*) pose environmental issues due to the generation of by-products and harmful substances.[5–7] In recent years, neutral hydrolysis has received widespread attention due to its superiority in terms of single products, environmental friendliness and renewability to polyester.[8–10] Liu *et al.* used zinc acetate as a catalyst to study the depolymerization reaction of PET in water, achieving a terephthalic acid (TPA) yield of 90% at 240 °C and 3.2 MPa after 30 min.[11] Yan *et al.* employed a mesoporous γ-Al_2O_3 supported Ni catalyst (Ni/γ-Al_2O_3) to catalyze the PET hydrolysis reaction at 267 °C, achieving a TPA yield of 97%.[12] Even though PET degradation can occur under neutral conditions, the energy consumption associated with high temperature and pressure poses a hindrance to its advancement. Hence, it is imperative to devise catalysts that can recycle and upgrade waste PET under gentle conditions.

Zeolites, characterized by their adjustable acid sites and well-defined microporous channels, have found extensive application in the upgrading and recycling of PET.[13–16] Recent research has shown that regulating the internal acid sites of zeolites can significantly promote the hydrolysis process of PET.[17] However, adjusting the structure of the zeolite itself is insufficient for achieving a substantial increase in acid sites. Introducing special active components, such as rare earth elements, represents a potential strategy to regulate the acid sites of zeolites, and related research has been applied in the degradation processes of plastics. Wu *et al.* found that incorporating the Ce element increases the acid site density of zeolites, significantly improving the selectivity of liquid fuels in polyethylene plastic degradation.[18] Wang *et al.* found that Ce-modified b-ZSM-5 exhibited a higher conversion rate in the hydrocracking of low-density polyethylene (LDPE) due to its increased acid site density.[19] Hence, we believe that utilizing rare earth elements to regulate the acidic properties of zeolites is a viable option for enhancing the PET hydrolysis activity. However, there have been limited relevant reports on this research.

Herein, we synthesized a Ce-modified HY zeolite to achieve the hydrolysis of PET under mild and neutral conditions, achieving a TPA yield of 93% at 190 °C in 3 h, which is 2.3 times higher than that of the HY zeolite. We further coupled

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Pd nanoparticles (NPs) with Ce-modified HY zeolites and found that the total hydrogenation yield reached 46.7% at 200 °C in 12 h with 3 MPa H_2. Our investigations point out that the introduction of Ce species enhances the amount of acid sites in the zeolite, which promotes the hydrolysis of PET and the upcycling of the product.

Results and discussion

Materials characterization

Y zeolites are synthesized using the hydrothermal method reported in previous literature.[20] Then, HY and *x*Ce-HY zeolites (where *x* denotes the loading amount of Ce) are obtained by an ion-exchange method (Fig. 1a and Table S1). Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images illustrate that the synthesized HY zeolites exhibit regular crystals and a smooth surface with a uniform size of approximately 300 nm (Fig. S1). As shown in Fig. 1b and c, when Ce is introduced, the HY zeolite remains as a well-ordered crystal, with no metal oxide particles observed on its surface. Furthermore, the energy-dispersive X-ray (EDX) mapping reveals a uniform distribution of Ce elements throughout the entire particle (Fig. 1d). In Fig. 2a, X-ray diffraction (XRD) patterns of HY, 1Ce-HY, 3Ce-HY and 5Ce-HY reveal characteristic diffraction peaks of a typical HY structure. In addition, the diffraction peaks located at 11.8° and 12.4° correspond to the (311) and (222) facets of the HY zeolite, respectively, and they are generally associated with the positioning of rare earth elements within the pores of the zeolite (Fig. 2b).[21–25] After Ce species are introduced, the peak at 11.8° in 1Ce-HY, 3Ce-HY and 5Ce-HY gradually decreases, which proves that Ce species enter the SOD cage of the HY zeolite. A new peak emerges at 12.4° in 5Ce-HY, which indicates that Ce species occupy the supercages of HY zeolites. These results

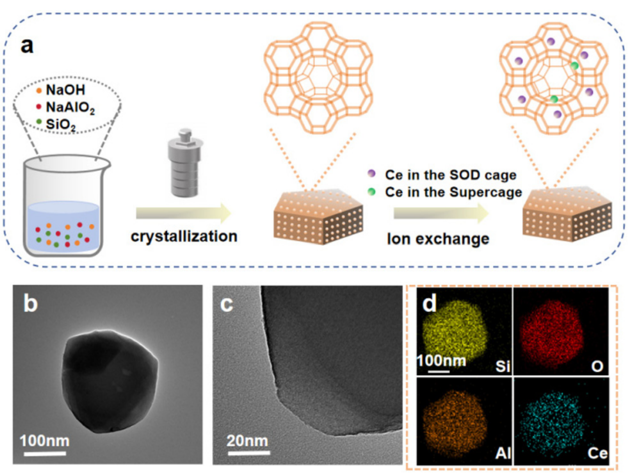

Fig. 1 Synthesis process and morphological characterization of HY and Ce-HY. (a) Schematic illustration of the synthesis of Ce-HY. (b) TEM image of 5Ce-HY. (c) HR-TEM image of 5Ce-HY. (d) EDX-mapping analysis of 5Ce-HY.

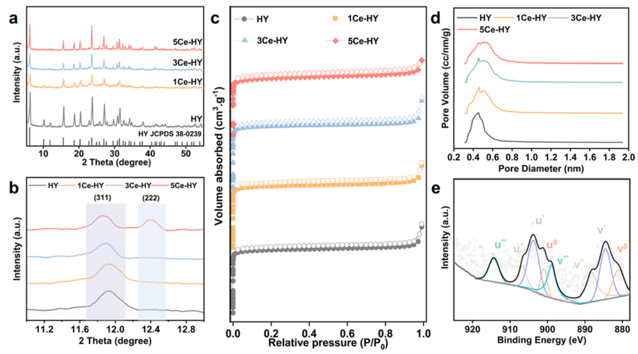

Fig. 2 Structural characterization. (a) XRD patterns, (b) amplified XRD patterns of 2 theta between 11.0° and 13.0°, (c) N_2 adsorption–desorption isotherms, and (d) pore size distributions of HY, 1Ce-HY, 3Ce-HY, and 5Ce-HY. (e) Ce 3d XPS spectra of the used 5Ce-HY.

indicate that the zeolite structure remains uncompromised upon the introduction of Ce species, which are successfully intercalated into the pores of the zeolite. N_2 adsorption–desorption measurements were carried out to get more information of the Ce-modified HY zeolites, and all the catalysts exhibited microporous structures. The Brunauer–Emmett–Teller (BET) surface areas of HY, 1Ce-HY, 3Ce-HY and 5Ce-HY are 989.52, 895.01, 876.31 and 860.35 m^2 g^{-1}, respectively, and the corresponding pore sizes are 0.30, 0.33, 0.34 and 0.33 nm, respectively. These results reveal that the Ce-modified HY zeolites have smaller surface areas and larger pore sizes than the HY zeolite, indicating that Ce species occupy the internal pores of zeolites (Fig. 2c and d).

To verify the surface state of Ce species, we performed X-ray photoelectron spectroscopy (XPS). As shown in Fig. 2e, the peaks located at V^1 (884.3 eV) and U^1 (903.7 eV) are attributed to Ce^{3+}, and the peaks located at V (881.2 eV), V^2 (888.1 eV), V^3 (898.8 eV), U (900.7 eV), U^2 (906.7 eV), and U^3 (914.4 eV) are attributed to Ce^{4+}. By integrating the different components of the Ce 3d spectrum, the ratio of Ce^{3+} in 5Ce-HY is found to be 0.58, indicating that the majority of the Ce elements are present in +3 valence.[26,27]

Furthermore, we loaded Pd species onto the surface of *x*Ce-HY zeolites using the impregnation method. As shown in Fig. S2a and S2b, distinct nanoparticles were observed on the surface of the 5Ce-HY zeolite, with the lattice spacing of 0.22 nm attributable to the (111) facet of metallic Pd, indicating successful loading of Pd. The energy-dispersive X-ray spectroscopy (EDX) mapping confirms the homogeneous distribution of the Pd element and Ce element across the entire particle (Fig. S2c). As shown in Fig. S3 and S4, the XRD pattern of Pd/*x*Ce-HY exhibits the characteristic peaks of the HY zeolite, as well as peaks indicative of Ce species, indicating that the synthesis process neither disrupts the framework structure of the zeolite nor alters the location of Ce species. Meanwhile, we observed diffraction peaks of metallic Pd, which are consistent with the TEM results. In the case of the Pd/HY catalyst, it is evident that metallic Pd nanoparticles are successfully loaded, while the structural integrity of the zeolite framework remains unaffected (Fig. S4 and S5).

Catalytic performance evaluation

The schematic of PET upgrading and recycling using the Ce-HY zeolite is shown in Fig. S6. The PET hydrolysis performance was evaluated at ambient pressure with a catalyst-to-PET mass ratio of 1 : 2 (100 mg of PET and 50 mg of catalyst). The influences of temperature and reaction time on PET hydrolysis were first investigated. As shown in Fig. 3a, when the reaction temperature is increased from 170 °C to 190 °C, the TPA yield of 5Ce-HY increases from 29% to 93%, indicating that the higher reaction temperature has a significant effect on the PET hydrolysis reaction. In terms of the impact of reaction time, the TPA yield of 5Ce-HY exhibits a positive correlation with the duration of the reaction, reaching a peak after 3 h (Fig. 3b). According to the above experimental results, the optimal reaction conditions are 190 °C for 3 h, reaching a TPA yield of 93%. For better comparison, we evaluated the catalytic performance of Ce-modified HY zeolite catalysts containing varying amounts of Ce. As shown in Fig. 3c and S7, the TPA yield of HY is only 39%, which increases to 51% with 1Ce-HY. As the Ce content rises, the yield of TPA gradually increases, peaking at 93% for 5Ce-HY, which is 2.3 times higher than that of HY. Furthermore, the cycling properties of 5Ce-HY in the PET hydrolysis reaction were also tested. As shown in Fig. S8, after three cycles of reaction, the TPA yield of 5Ce-HY decreases to 28%. The XRD pattern reveals that after three cycles of reaction, the 5Ce-HY catalyst still retains the characteristic diffraction peaks of the HY zeolite and the Ce species remain located within the supercage and SOD cage. However, the decrease in the intensity of the diffraction peaks indicates a reduction in crystallinity (Fig. S9 and S10). In addition, NH_3-TPD results revealed a marked decrease in the signal intensity of 5Ce-HY after reaction. This indicates that the reduction in crystallinity destroyed some acid sites, leading to a decline in catalytic activity (Fig. S11).[28] We also compared the performance of different zeolites, with 5Ce-HY demonstrating the most outstanding performance (Fig. S12).

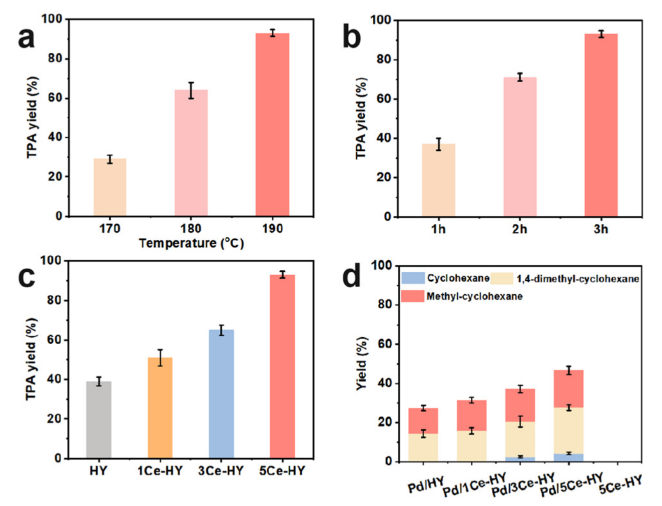

Fig. 3 Catalytic performances. (a–c) Catalytic performances in PET hydrolysis. (a) Yields of TPA for 5Ce-HY at different temperatures in 3 h. (b) Yields of TPA for 5Ce-HY at 190 °C at different times. (c) Catalytic performance of PET hydrolysis over different catalysts. Reaction conditions: 100 mg of PET, 50 mg of catalyst, 10 mL of water, 190 °C, and 3 h. (d) Catalytic performance of TPA hydrogenation over different catalysts. Reaction conditions: 100 mg of TPA, 50 mg of catalyst, 10 mL of *n*-hexane, 200 °C, feed gas at 3 MPa with H_2, and 12 h. The error bar is the standard deviation of three independent measurements. The data are displayed as the mean ± SD (n = 3).

Furthermore, to achieve upcycling of PET, we investigated the hydrogenation performance of TPA. The hydrogenation reaction was conducted at 200 °C for 12 h, using *n*-hexane as the solvent, with a mass ratio of catalyst to TPA of 1 : 2 (100 mg of TPA and 50 mg of catalyst). As shown in Fig. 3d, the conversion of TPA over the 5Ce-HY catalyst is nearly negligible, potentially due to the poor activation ability of the zeolite towards H_2. Consequently, we incorporated Pd species into the *x*Ce-HY catalyst (Pd/*x*Ce-HY). First, the effects of reaction temperature and reaction pressure on the TPA hydrogenation reaction were investigated. As shown in Fig. S13, the TPA conversion increases gradually with an increase in reaction temperature, reaching a peak value of 46.7% at 200 °C. Under atmospheric pressure, the hydrogenation reaction of TPA barely proceeds. However, when the reaction pressure is increased from 0.1 MPa to 3 MPa, the TPA hydrogenation yield of Pd/5Ce-HY increases from 0% to 46.7%, indicating that the H_2 pressure is vital to the TPA hydrogenation process (Fig. S14). For better comparison, the performance of catalysts with different Ce contents was evaluated. As shown in Fig. 3d and Fig. S15, under the optimal reaction conditions, the TPA hydrogenation yield of Pd/HY is 27.3%, with the main products being methyl-cyclohexane and 1,4-dimethyl-cyclohexane. However, with the introduction of Ce species, the hydrogenation yield increases to 31.4% for Pd/1Ce-HY. As the Ce content increases, the yields of TPA hydrogenation products gradually increase, peaking at 46.7% for Pd/5Ce-HY. Meanwhile, methyl-cyclohexane and 1,4-dimethyl-cyclohexane are the main products, accompanied by a small amount of cyclohexane. Additionally, we explored the effect of metal selection. Pd emerged as the optimal choice, exhibiting exceptional TPA hydrogenation activity with economic benefits (Fig. S16). We further verified the stability of Pd/5Ce-HY. As shown in Fig. S17 and Table S2, with an increase in the number of cycles, the performance of Pd/5Ce-HY gradually declines, which may be attributed to the partial leaching of the metal into the reaction solution.[29]

Mechanism investigation

To investigate the catalytic mechanism, we characterized the acidic properties of the catalysts using NH_3-temperature-programmed-desorption (NH_3-TPD). As shown in Fig. 4a, there are two peaks in the NH_3-TPD profiles, situated at 140 °C and 310 °C, which are assigned to the adsorption of ammonia at the weak acid sites and the medium strength acid sites, respectively.[30,31] With an increase in Ce content, there is a notable intensification in the desorption peak intensity,

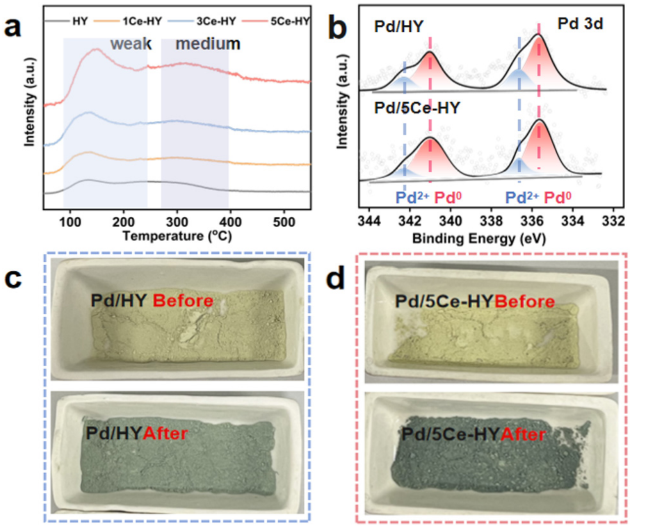

Fig. 4 (a) NH_3-TPD profiles of HY and Ce-HY. (b) Pd 3d XPS spectra of the used Pd/HY and Pd/5Ce-HY. (c and d) Photographs of H-spillover detected using WO_3. Samples were prepared with 1 g of WO_3 mixed with 0.02 g of various catalysts after treatment with H_2 at 100 °C for 10 min.

accompanied by a shift towards higher temperatures, suggesting that the Ce species enhance the amount and strength of acid sites. Furthermore, we synthesized a series of RE (rare earth element)-incorporated HY catalysts. The PET hydrolysis performance aligned with the NH_3-TPD signal changes, proving the pivotal role of acid sites in catalytic performance (Fig. S18 and S19).

To verify the hydrogenation reaction mechanism, we conducted XPS spectra to explore the surface states of the catalyst. As shown in Fig. 4b, the peaks near 335.6 eV and 340.9 eV are attributed to Pd^0, while those near 336.6 eV and 342.2 eV are attributed to Pd^{2+}. It can be observed that there is no shift in the binding energy of Pd after the introduction of Ce species. Additionally, the Ce 3d XPS spectra reveal that Pd/5Ce-HY possesses binding energies comparable to those of 5Ce-HY, suggesting the absence of coupling between the Pd and Ce species (Fig. S20). As evidenced by the previously mentioned TEM, EDX mapping, and XRD results, the incorporation of Pd nanoparticles does not alter the structural integrity of the zeolite or disrupt the positioning of Ce within the zeolite pores (Fig. S2–S4). Furthermore, considering the similar Pd loading of the catalysts and the positive correlation between Ce loading and hydrogenation yield, it is suggested that Ce is a critical factor influencing the hydrogenation behavior.

To investigate the pivotal role of Ce species, we utilized WO_3 as an indicator to probe the H-spillover effect (Fig. 4c and d). WO_3 was physically mixed with the various catalysts in a ratio of 50 : 1, and the color change of the compound was observed after exposing it to H_2. After 10 min of H_2 gas injection at 100 °C, the degree of color change observed in the mixture of Pd/5Ce-HY and WO_3 is greater compared to that observed in the mixture of Pd/HY and WO_3, indicating that the Pd/5Ce-HY catalyst exhibited the strongest H-spillover effect. This significant spillover effect may originate from the increase in acid sites within the zeolite caused by the introduction of Ce species, which is consistent with previous reports. The enhanced H-spillover effect of the corresponding Pd/5Ce-HY can effectively promote the hydrogenation process.[32–36]

Conclusions

In summary, we have developed a strategy for hydrolyzing PET under mild conditions by modifying the HY zeolite with the Ce element. The experimental and mechanism studies reveal that the introduction of Ce significantly enhances the content and strength of acid sites within the zeolite, which effectively promotes the hydrolysis of PET. Furthermore, the acid sites facilitate the H-spillover effect, which is beneficial for the hydrogenation process of the hydrolysis product (TPA). This work provides a feasible strategy for catalyst design in hydrolysis and hydrogenation degradation experiments of PET. We anticipate that rare-earth-modified zeolites will facilitate the upgrading of waste plastics and be used for a wider range of catalytic processes.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the SI. Supplementary information: figures showing the characterization of the as-prepared nanoparticles. See DOI: **https://doi.org/10.1039/d5dt01089f**.

Acknowledgements

This work was supported by the financial aid from the National Science and Technology Major Project of China (2021YFB3500700), the National Natural Science Foundation of China (22020102003, 22025506, 22271274 and U23A20140) and the Jilin Province Science and Technology Development Plan Project (20240402056GH). X. W. acknowledges funding from the National Natural Science Foundation of China Outstanding Youth Science Foundation of China (Overseas).

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